Central Assessment of Longitudinal Knee X-rays for Femoral-Tibial Angle (Anatomic Alignment)

1. Overview 1
1.1 SAS dataset 1
1.2 Contents of dataset 1
1.3 Condition 2
1.4 Variables and reading methods 2
2. Methods specific to Project 17 5
2.1 Image type 5
2.2 Time points 5
2.3 Measurement methods 5
2.4 Variables 5
2.5 Sample 5
3. References 6

1. Overview

1.1 SAS dataset

Name: **kXR_FTA_DuryeaXX** (XX identifies the time point)
Display label: **kXR FTA (Duryea)**

1.2 Contents of dataset

This dataset contains central baseline visit measurements of femoral-tibial angle (anatomic axis alignment) from serial OAI knee x-rays, performed in the laboratory of Dr. Jeff Duryea at Brigham and Women's Hospital in Boston, MA. http://www.spl.harvard.edu/pages/People/duryea. The data from more than one project by this vendor may be contained in the dataset. A "reading project" consists of a particular sample of knees read over a specific period of time using a single protocol.

NOTE: In this release of data (versions 0.3, 1.2, 3.2, 5.2, 6.2, 8.2, and 10.2), all OAI participants with longitudinal knee x-rays who have knee OA, along with a large number of participants without knee OA and the entire non-exposed control subcohort of OAI have FTA measurements from their baseline and follow-up visit knee x-rays provided. Compared to the previous release of these datasets, additional measurements on 267 additional knees have been provided – almost all knees which do not have radiographic OA at any time point. In general any knee that has a quantitative JSW measurement in the kXR_QJSW_DuryeaXX datasets will also have a measurement of FTA in these datasets at the same time points.

Each dataset contains the reading data for a single visit, or time point. The corresponding longitudinal data from reading projects by this vendor for other time points are in separate datasets. For example, the data from this vendor's reading of baseline x-rays are in a dataset ending in "00" (i.e., kXR_FTA_Duryea00), while the corresponding data from this vendor for the 12-month visit x-ray readings are in a dataset with the same name but ending in "01" (i.e., kXR_FTA_Duryea01), and so on. See the "*VisitPrefixDefinitions.pdf*" document for a guide to visit numbering.) To compare values of a variable across time points from a given project by this vendor, or to calculate change scores, users will need to merge the datasets for the various time points.

IMPORTANT: **All users are strongly encouraged to read the *Overview and Description of Central Image Assessments* document carefully before attempting to use this dataset!**

For comprehensive and detailed information about the structure and contents of the central image assessment datasets, general guidelines for defining change between time points and duration of follow-up, and strategies for merging datasets for analysis, please see the "*Overview and Description of Central Image Assessments*" document.

1.3 Condition

- Known data errors: problems/cautions for use are listed by variable in the "Release Comments", which can be found in the various *kXR_FTA_DuryeaXX_Comments.pdf* files.

- Dataset strengths/weaknesses:
  - Data are expected for all participants included in a project sample. If expected data do not exist for a knee, SAS special missing values are assigned to denote why the data were not obtained.
  - The dataset only contains a single row of data (1 record) for a given knee, but other OAI datasets may contain multiple records per knee and this needs to be taken into account when merging with other datasets. Please see the "*Overview and Description of Central Image Assessments*" document merging.

  - The measurement of FTA requires that at least 10cm of the tibia is visible and in some knee radiographs this is not true. A flag variable has been provided to indicate when this has occurred.
- Overlap between reading projects in this dataset:
  - The knees with baseline visit FTA measurements in this dataset are also likely to have JSW measurements provided from the same x-rays in the JSW x-ray reading datasets (Project 16, kXR_QJSW_DuryeaXX datasets).

1.4 Variables and reading methods

See the dataset documentation file *kXR_FTA_DuryeaXX_Contents.pdf* for a complete list of all the variables in the dataset, their SAS variable names, descriptive variable labels and attributes.

Key variables measured for all projects by this vendor include:

- FTA measurements (in degrees), where increasing FTA indicate more valgus malalignment.
- A flag variable to indicate if enough of the tibial shaft (10cm) was visible on the x-ray for reliable assessment of the tibial shaft axis direction.

The image analysis, measurement techniques and variables are similar in the different projects by this vendor, and are outlined below.

Any modifications for a specific project and project specific variables are described in the section related to that project, including:

- Measurements made at each time point; and
- Criteria for selecting the knees to analyze

The measurement of FTA involves the definition of the femoral axis using a coordinate system based on the shape of the femoral condyles that is defined as part of the location specific joint space width measurement methods used by Dr. Duryea[1,2]. Unlike measurement of malalignment from the mechanical axes of the leg from full-limb radiographs using HKA (hip-knee-ankle angle) – see OAI Project 60 – where neutral alignment is represented by HKA=0 degrees, a FTA value of around -4.7 degrees typically exists in limbs which are neutrally aligned on full-limb radiographs[3]. As is the convention with HKA, for the FTA measurements, knees with valgus malalignment have more +ve FTA values than knees with varus malalignment.

These publications give more details about the methods and measurement performance:

- *Neumann G, Hunter D, Nevitt M, Chibnik LB, Kwoh K, Chen H, Harris T, Satterfield S, Duryea J, For the Health ABC Study. Location specific radiographic joint space width for osteoarthritis progression. Osteoarthritis Cartilage. 2009 Jun; 17(6): 761-765. PMID: 19073368* http://dx.doi.org/10.1016/j.joca.2008.11.001
- *Duryea J, Li J, Peterfy CG, Gordon C, Genant HK. Trainable rule-based algorithm for the measurement of joint space width in digital radiographic images of the knee. Med Phys. 2000 Mar;27(3):580-91. PMID: 10757609* http://dx.doi.org/10.1118/1.598897
- *Iranpour-Boroujeni T, Li J, Lynch J, Nevitt M, Duryea J. A new method to measure anatomic knee alignment for large studies of OA: data from the Osteoarthritis Initiative. Osteoarthritis and Cartilage 2014 Oct; 22(10): 1668-1674 PMID: 25278076* http://dx.doi.org/10.1016/j.joca.2014.06.011

In general, a customized software tool developed by Dr. Duryea is used to provide a measurement of JSW and FTA on serial fixed-flexion knee radiographs from the OAI. Figure 1 shows the definition of the femoral and tibial axes in the software.

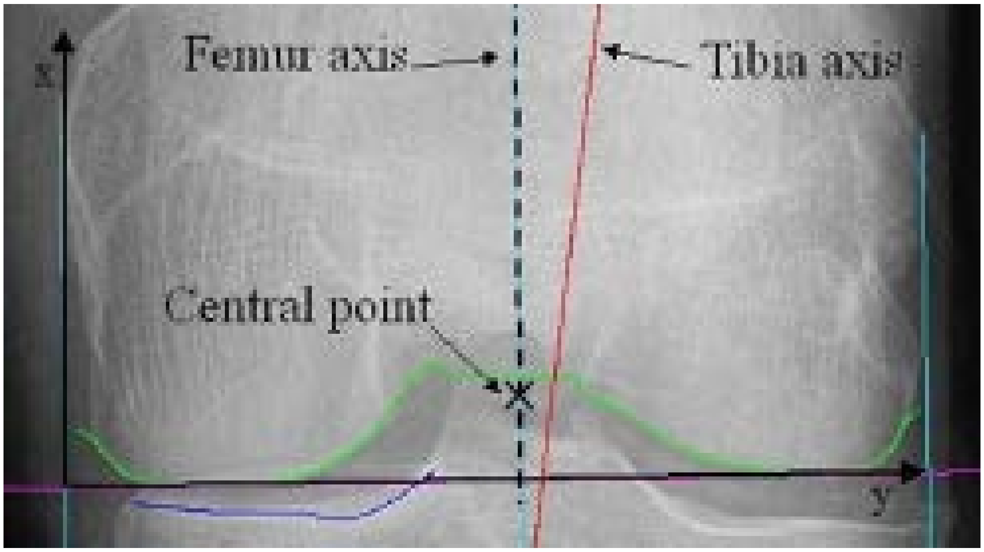

Figure 1. Output of the software on a digital knee radiograph. The femoral axis if defined from the y-axis of the location specific JSW measurement procedure. The tibial shaft axis is defined based on the direction of the tibial shaft at a distance of 10cm below the tibial plateau.

The tibial axis is usually defined by the direction and center of the tibial shaft at a distance 10cm below the tibial plateau. In some radiographs, the knee position meant that less than 10cm of the tibia was visible. A flag variable is provided in the dataset to indicate which knees are affected by this issue. FTA measurements for these knees may not be reliable and depending on the analysis method being used, this should be taken into account when data are analyzed.

Missing data can occur for a variety of reasons. In the individual kXR_FTA_DuryeaXX datasets, SAS special missing values are used to indicate the following:

.P if data is missing due to a prosthesis/knee replacement

.T if data is missing due to technical reasons

.A if the data is not expected

2. <u>Methods specific to Project 17</u>

2.1 Image type:
Fixed-flexion bilateral knee radiograph. See the "*Radiographic (X-ray) Manual*" document for the acquisition protocol.

2.2 Time points:
Baseline, 12-month, 24-month, 36-month, 48-month, 72-month, and 96-month visits.

2.3 Measurement methods:
The manual and semi-automated aspects of the reading were done viewing the images paired but blinded to chronological order, with paired images viewed simultaneously.

2.4 Variables:
For this project:

- FTA measurements (in degrees), where increasing FTA indicate more valgus malalignment.
- A flag variable to indicate if enough of the tibial shaft (10cm) was visible on the x-ray for reliable assessment of the tibial shaft axis direction.

2.5 Sample:
See section 1.2 for a description of the sample. The following table gives some demographic information about the participants with data currently released for Project 17:

Project 17 sample: Distribution of Race by Sex

	White or Caucasian	Non-White	Total
Male	1,185	225	1,410
Female	1,529	495	2,024
Total	2,714	720	3,434*

*Race data missing for 1 participant

Project 17 sample: Distribution of Age by Sex

	Age (years)				
	45 to 49	50 to 59	60 to 69	70 to 79	Total
Male	161	508	370	372	1,411
Female	194	657	708	465	2,024
Total	355	1,165	1,078	837	3,435

3. References

1. Neumann G, Hunter D, Nevitt M, Chibnik LB, Kwoh K, Chen H, Harris T, Satterfield S, Duryea J, For the Health ABC Study. Location specific radiographic joint space width for osteoarthritis progression. Osteoarthritis Cartilage. 2009 Jun; 17(6): 761-765. PMID: 19073368 http://dx.doi.org/10.1016/j.joca.2008.11.001
2. Duryea J, Li J, Peterfy CG, Gordon C, Genant HK. Trainable rule-based algorithm for the measurement of joint space width in digital radiographic images of the knee. Med Phys. 2000 Mar;27(3):580-91. PMID: 10757609 http://dx.doi.org/10.1118/1.598897
3. Iranpour-Boroujeni T, Li J, Lynch J, Nevitt M, Duryea J. A new method to measure anatomic knee alignment for large studies of OA: data from the Osteoarthritis Initiative. Osteoarthritis and Cartilage 2014 Oct; 22(10): 1668-1674 PMID: 25278076 http://dx.doi.org/10.1016/j.joca.2014.06.011